GyaanConnect

STABILIZATION SPRINT SPECIFICATION

Audit-Driven Stabilization Sprint — Step 1 Verification & Proposed Plan

Architecture Preservation: React 19 • Vite 8 • Tailwind CSS v4 • Node.js ESM HTTP Server (Port 3001)

SPRINT PHASE	PRIMARY GOAL	PRE-CONDITION	TESTING FRAMEWORK
Phase 1: Stabilization	Security & Codebase Health	Zero Rebuild / No Mock Masquerade	Vitest Automated Harness

Step 1: Independent Verification of the 10 Audit Findings

Prior to making any file changes, every finding was independently verified against the current repository source code:

#	Audit Finding	Status	Verified Code Reference & Technical Evidence
1	`npm run lint` failure due to `.vite_cache`	CONFIRMED	`vite.config.js#L7` sets `cacheDir: './.vite_cache'`. Oxlint scans vendor code in `.vite_cache/deps/`, triggering **3,186 warnings and 14 errors** (exit code 1).
2	Missing curriculum API routes	CONFIRMED	`curriculumService.js#L32,L48` fetches `/api/curriculum/subjects` and `/api/curriculum/chapters`. Neither route exists in `server/index.js` router, returning **HTTP 404**.
3	`POST /api/evaluate` trusts client rubric	CONFIRMED	`server/index.js#L31-L60` builds prompt using `targetConcept.mustKnowPoints` directly from client request body. Vulnerable to prompt injection cheat scoring.
4	API keys persisted in `localStorage`	CONFIRMED	`GeminiSettingsModal.jsx#L22` and `App.jsx#L98` save keys to `localStorage.setItem('gyan_gemini_key', key)`, exposing credentials to browser extensions.
5	Direct client-to-Gemini browser calls	CONFIRMED	`aiVoiceService.js#L261-L275,L472-L484` executes raw `fetch()` from browser to `https://generativelanguage.googleapis.com/...` using the client-stored key.
6	Conflicting Board definitions across files	CONFIRMED	`conceptKnowledgeBase.js` has 9 boards, `curriculumDatabase.js` has 5, and `curriculumData.js` has 3. `curriculumData.js#L374` defaults any non-CBSE/ICSE board to UP Board.
7	Procedural fake chapter generation	CONFIRMED	`curriculumData.js#L647-L727` runs `generateChaptersForSubject()` synthesizing fake titles ("*Chapter 1: Mathematics - Unit 1*") and generic quizzes for unverified classes.
8	Automated test suite is missing	CONFIRMED	`package.json` contains no `test` script, no test runner (Vitest/Jest) is installed, and zero `*.test.js` or `*.spec.jsx` files exist in the project.
9	React state-update warnings in effects	CONFIRMED	`npx oxlint src server` confirms 2 warnings: synchronous `setLoading(true)` in `ConceptSelectionGrid.jsx#L37` and `setPhase('briefing')` in `ProveUnderstandingStudio.jsx#L76`.
10	`AiVoiceRoom.jsx` is orphaned dead code	CONFIRMED	Grep search confirms `src/components/AiVoiceRoom.jsx` (584 lines) is not imported or referenced anywhere. It was superseded by `ProveUnderstandingStudio.jsx`.

Step 2: Phased Implementation Work Orders (A → G)

WORK ORDER A **Fix Lint Exclusions & React Effect Warnings**

- Add `.vite_cache` and `dist` to `.gitignore` and configure `.oxlintrc.json` ignore patterns.
- Update `package.json` lint script to `"oxlint src server"` as a dual-layer guard.
- In `ConceptSelectionGrid.jsx`: remove synchronous `setLoading(true)` on effect entry; manage loading via async dispatch.
- In `ProveUnderstandingStudio.jsx`: remove the 7 synchronous state resets on concept change; apply React 19 `key={selectedConcept?.id}` in `LectureTheater.jsx` to reset state cleanly without cascading re-renders.
- *Acceptance Criteria:* `npm run lint` passes with **0 errors and 0 warnings**.

WORK ORDER B **Secure AI Evaluation via Server-Side Rubric Resolution**

- Refactor `handleEvaluate` in `server/index.js`: require valid `conceptId` and `studentAnswer` (≥ 5 characters).
- Query authoritative concept in `server/curriculumDatabase.js` using `conceptId`. Return HTTP 400/404 if missing or invalid.
- **Strict Boundary:** Never read `mustKnowPoints` or `commonMisconceptions` from client payload. Inject only server-side authoritative criteria into the prompt.
- Apply authoritative resolution and answer validation to `handleFollowUp` (`POST /api/evaluate/followup`).
- If server API key is unconfigured or Gemini times out, return HTTP 503 with `{ fallbackToLocal: true }`.

WORK ORDER C **Remove Client API Key Storage & Direct Gemini Requests**

- In `src/services/aiVoiceService.js`: completely remove `callGeminiDirect()` and direct browser `fetch()` calls.
- Routing becomes: *Browser Explains → Backend Proxy (`/api/evaluate`) → (fallback) Local Offline Socratic Engine.*
- In `src/components/GeminiSettingsModal.jsx`: remove raw key input; rebrand into a secure **"AI Diagnostic & Voice Settings"** modal displaying server connection status, child privacy assurance, and TTS voice settings.
- Remove `localStorage.getItem('gyan_gemini_key')` and `localStorage.setItem('gyan_gemini_key')` across all components.

WORK ORDER D **Implement Missing Curriculum Endpoints**

- Add `GET /api/curriculum/subjects` in `server/index.js` with `board` and `class` parameter validation (1–12).
- Add `GET /api/curriculum/chapters` in `server/index.js` with `board`, `class`, and `subject` parameter validation.
- Return authentic subjects and verified chapters. For in-development topics, return accurate metadata marked `status: "curation_pending"` instead of generating fake unit numbers.

WORK ORDER E **Unify Board Schema & Eliminate Chapter Fabrication**

- Consolidate `BOARDS` into a single unified schema across frontend and backend (CBSE, ICSE, UP Board, and State Boards).
- Ensure selecting any board in `curriculumData.js` properly matches without defaulting to UP Board.
- In `curriculumData.js`: replace `generateChaptersForSubject()` procedural loops with authentic NCERT Rationalised syllabus chapter titles marked `status: 'curation_pending'`, eliminating fake repetitive quiz questions.

WORK ORDER F **Safely Remove Orphaned Dead Code**

- Confirm zero references, delete `src/components/AiVoiceRoom.jsx` (584 lines), and verify `npm run build` succeeds cleanly.

WORK ORDER G **Vitest Automated Test Harness**

- Install `vitest` and add `"test": "vitest run"` to `package.json`.
- Implement 5 automated test suites:
  1. `tests/conceptGraphEngine.test.js`: DAG acyclic validation, prerequisite health, SM-2 interval calculations.
  2. `tests/scoring.test.js`: local evaluation scoring bounds (0–100), must-know coverage, misconception penalties.
  3. `tests/rubricSecurity.test.js`: reject invalid concept IDs; verify server ignores client-injected rubrics.
  4. `tests/curriculumEndpoints.test.js`: verify `/boards`, `/classes`, `/subjects`, `/chapters`, and `/concepts`.
  5. `tests/aiResponseValidation.test.js`: test `normalizeEvaluationResponse` against malformed or partial JSON.

Step 3: Feature Preservation & Regression Guard

Existing Feature	Core Preservation Mechanism	Regression Test Method
Feynman Oral Recall Studio	Retain 4-phase state machine, confetti triggers, and audio recording controls.	Verify oral proof UI and 5-pillar evaluation modal.
Voice & Speech (Hindi / English)	Preserve `SpeechManager` in `aiVoiceService.js` with language switching.	Test speech recognition and TTS playback controls.
Offline Socratic Evaluation	Preserve `evaluateLocally()` heuristic engine as fallback when server has no key.	Automated unit tests in `scoring.test.js`.
Concept Graph Lineage	Preserve DAG node visualizer and prerequisite health checks.	Automated tests in `conceptGraphEngine.test.js`.
Spaced Repetition Queue	Preserve SM-2 interval scheduler (1d, 3d, 7d) and overdue task queries.	Unit tests for interval calculations and dashboard display.
Lecture Theater & Quizzes	Preserve video marathon player, practice quizzes, and XP allocation.	Verify subject switching and quiz scoring.
Profile Backups & Global Search	Preserve JSON export/import functions and Ctrl+K search modal.	Test file download/upload and search filter results.

Step 4: Verification Protocol

1. **Unit Testing:** Run `npm test` and verify 100% test pass rate across all 5 test suites.
2. **Linter Verification:** Run `npm run lint` and confirm 0 errors and 0 warnings.
3. **Production Build Verification:** Run `npm run build` and confirm clean bundle generation.
4. **API Health & Security:** Boot `node server/index.js` and verify health, curriculum routes, and rejection of invalid `conceptId`.
5. **Browser Testing:** Test voice recording, offline evaluation, and UI interactions in a Chromium browser.